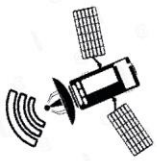
Aix*Marseille université
Socialement engagée

Departement Géographie, aménagement, environnement et développement - Parcours Géomatique et modélisation spatiale (GMS)

UEs : HGAAU39 & HGAAU38 & HGAAU37





Comparative analysis of the performance of NASA Landsat OLI/TIRS, ESA Sentinel 2 MSI, and Roskosmos Kanopé V satellites in Klaipeda (Lithuania)





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ACRONYMS AND ABBREVIATIONS

CLC	CORINE Land Cover
GS	Ground Segment / Band Name (Kanopé V)
Kanopé V	N/A
L8	Landsat 8
L8 LULC	Landsat 8 Land Use / Land Cover
LULC	Land Use and Land Cover
MNF	Minimum Noise Fraction
MSI	MultiSpectral Instrument
NIR	Near-Infrared
OLI	Operational Land Imager
OSM	OpenStreetMap
PCA	PrincipalComponent Analysis
ROI	Region of Interest
S2	Sentinel-2
SWIR	Shortwave Infrared
SWIR B11	Shortwave Infrared Band 11
TIRS	Thermal Infrared Sensor
Urban Atlas	N/A

ABSTRACT

This project focuses on the detection, recognition, and temporal analysis of land-cover and land-use objects in the city of Klaipėda, a coastal port city in western Lithuania, using multi-source satellite imagery. The study integrates very high resolution reference imagery (Kanopé V, November 2021) with medium-resolution Earth observation data to assess the capabilities and limitations of different satellite vectors for spatial analysis at the building and urban morphological unit scales.

In the first phase, a preliminary typology of land-cover objects, structures, and morphologies was established, including water bodies, agricultural fields, wooded areas, bare or sparsely vegetated soils, and built-up surfaces. Detectability levels were defined for each object type based on spatial resolution, spectral response, and morphological characteristics. Sentinel-2 MSI and Landsat 8/9 OLI-TIRS data were resampled and, where relevant, pansharpened to improve spatial coherence. A detectability matrix linking satellite vectors, spectral domains (visible, SWIR, thermal), and geographical object types was produced.

Several methodologies were developed to recognise and map land-use and land-cover classes, combining pixel-based supervised classification and Geographic Object-Based Image Analysis (GEOBIA). The choice of classification algorithms (Random Forest, Support Vector Machine, and rule-based approaches) was justified through comparative testing using representative samples. Classification results were validated using open reference databases such as Urban Atlas, OpenStreetMap, Lithuanian national open datasets, and available 3D data. Validation metrics were computed by object type, providing quantitative estimates of recognition, identification, and classification accuracy at the building and urban morphological unit scales.

The second part of the study investigated recent spatial changes in Klaipėda and its coastal environment from 2013 to 2025 using a harmonized time series of Sentinel-2 MSI and Landsat 8/9 OLI-TIRS imagery. Images were selected for consistent seasonal periods and resampled to common spatial resolutions (10 m for Sentinel-2 and 15 m for Landsat). One of the classification models developed in the first phase was adapted and applied to each date following a standardized methodological workflow. Change detection maps were produced to identify land-cover transitions and statistically assess trends using regression analysis. A comparative evaluation of Sentinel-2 and Landsat results highlighted their complementary strengths for urban and coastal monitoring over time.

Key words: Remote sensing, Urban morphology, Land-use classification, Coastal city, Urban dynamics

INTRODUCTION

Located in the far west of Lithuania, Klaipėda opens onto the Baltic Sea at the mouth of the Klaipėda Strait, which connects the Curonian Lagoon to the open sea. This coastal position places the city at the interface between continental and maritime environments and gives it a strategic role in exchanges between the European continent and the Baltic region. With approximately 150,000 inhabitants and covering nearly 98 km², Klaipėda is Lithuania's third largest city and the country's only seaport, making it a major urban, industrial, and logistical center whose development is closely linked to maritime activities.

The territory of Klaipėda exhibits strong spatial heterogeneity, reflecting the diversity of its urban functions and land-use patterns. The historical city center concentrates administrative, commercial, and cultural activities, while port and industrial areas extend primarily toward the southern and western sectors. Residential expansion has mainly occurred in the northern and eastern parts of the city, following recent urban growth dynamics. At the same time, natural environments such as coastal dunes, forests, wetlands, and lagoon margins form an ecological belt around the urban fabric, contributing to the city's environmental and landscape diversity.

Despite its strategic position and economic importance, Klaipėda faces significant environmental and economic challenges. Urban expansion and industrial activities exert pressure on natural and coastal ecosystems, while coastal erosion, sea-level rise, and increased storm intensity threaten both urban areas and port infrastructures. As Lithuania's only seaport and a key hub for maritime trade in the Baltic Sea, the city must also address challenges related to modernizing port facilities, adapting infrastructure to climate change, and reducing pollution from maritime and industrial traffic. In this context, the development of green logistics, energy transition strategies, alternative fuels, and circular economy practices represents a major challenge for achieving sustainable urban and coastal development.

I. PROBLEM STATEMENT

Coastal cities are subject to complex and rapidly evolving spatial dynamics driven by urban growth, economic development, and environmental pressures. In such environments, understanding land-cover and land-use transformations is essential for sustainable planning and risk management, particularly where urban, industrial, and natural systems coexist (Weng, 2012; Giri, 2012). Remote sensing offers powerful tools for observing these dynamics over large areas and long time periods, but the effectiveness of satellite-based monitoring depends strongly on sensor characteristics such as spatial resolution, spectral capabilities, and temporal coverage (Drusch et al., 2012; Roy et al., 2014). This raises methodological challenges when analyzing heterogeneous coastal urban territories (Blaschke, 2010).

Klaipėda, Lithuania's only seaport and a major economic gateway to the Baltic Sea, presents a particularly relevant case study for multi-sensor urban and coastal analysis. The city combines dense built-up areas, port and industrial infrastructures, agricultural fields, forests, wetlands, water bodies, and sensitive coastal environments within a relatively compact territory (Pesaresi et al., 2016; Congalton & Green, 2019). Monitoring changes in such a heterogeneous landscape over more than a decade requires data that can simultaneously capture fine-scale urban structures and broader land-cover dynamics. While medium-resolution sensors such as Landsat OLI/TIRS and Sentinel-2 MSI provide long-term, systematic observations, very high resolution imagery such as Kanopé V offers detailed spatial information that can serve as a reference for object detection and validation (Drusch et al., 2012; Roy et al., 2014). However, their respective capacities for identifying different types of geographical objects and land-use functions remain uneven and scale-dependent (Blaschke, 2010; Weng, 2012).

In this context, this study aims to assess and compare the capacities of multiple Earth observation sensors for urban and coastal monitoring by addressing the following research question: to what extent do the Landsat OLI/TIRS, Sentinel-2 MSI, and Kanopé V sensors enable the effective detection, identification, and monitoring of urban and coastal transformations in Klaipėda between 2013 and 2025, and what are their respective performances according to observation scales and the types of objects studied?

The results are expected to provide insights into the strengths and limitations of each sensor and to guide future applications of multi-sensor approaches for sustainable urban and coastal management.

II. DATA AND METHODOLOGY (FLOWCHART)

1. Data Sources

This study is based on three satellite sensors with complementary spatial and spectral characteristics. Landsat 8/9 OLI-TIRS imagery was used for the period 2013–2025, providing multispectral information in the visible, near-infrared, shortwave infrared, and thermal domains, with a panchromatic band enabling spatial enhancement. Sentinel-2 MSI data were used from 2015 to 2025, offering higher spatial resolution multispectral imagery, including visible, red-edge, near-infrared, and shortwave infrared bands well suited for urban and environmental analysis. Very high-resolution Kanopé V imagery acquired in November 2021 was used as a reference dataset for detailed observation of urban and coastal structures. Additional open spatial data, including building footprints and land-use layers, were used to support training and validation.

2. Image Processing (TerrSet)

All image processing was carried out using TerrSet software. For Landsat 8/9, pansharpening was applied to all multispectral bands using Band 8 as the spatial reference, enhancing the detection of urban elements such as built-up areas, road networks, and industrial structures. For Sentinel-2, SWIR and other lower-resolution bands were resampled and sharpened using Band 8 as the spatial reference, ensuring consistent spatial resolution across bands relevant for urban and land-cover analysis. No pansharpening or resampling was applied to Kanopé V imagery, as its very high spatial resolution was already sufficient for object-level analysis. These processing steps ensured spatial coherence between sensors while preserving spectral information relevant to urban and coastal feature extraction.

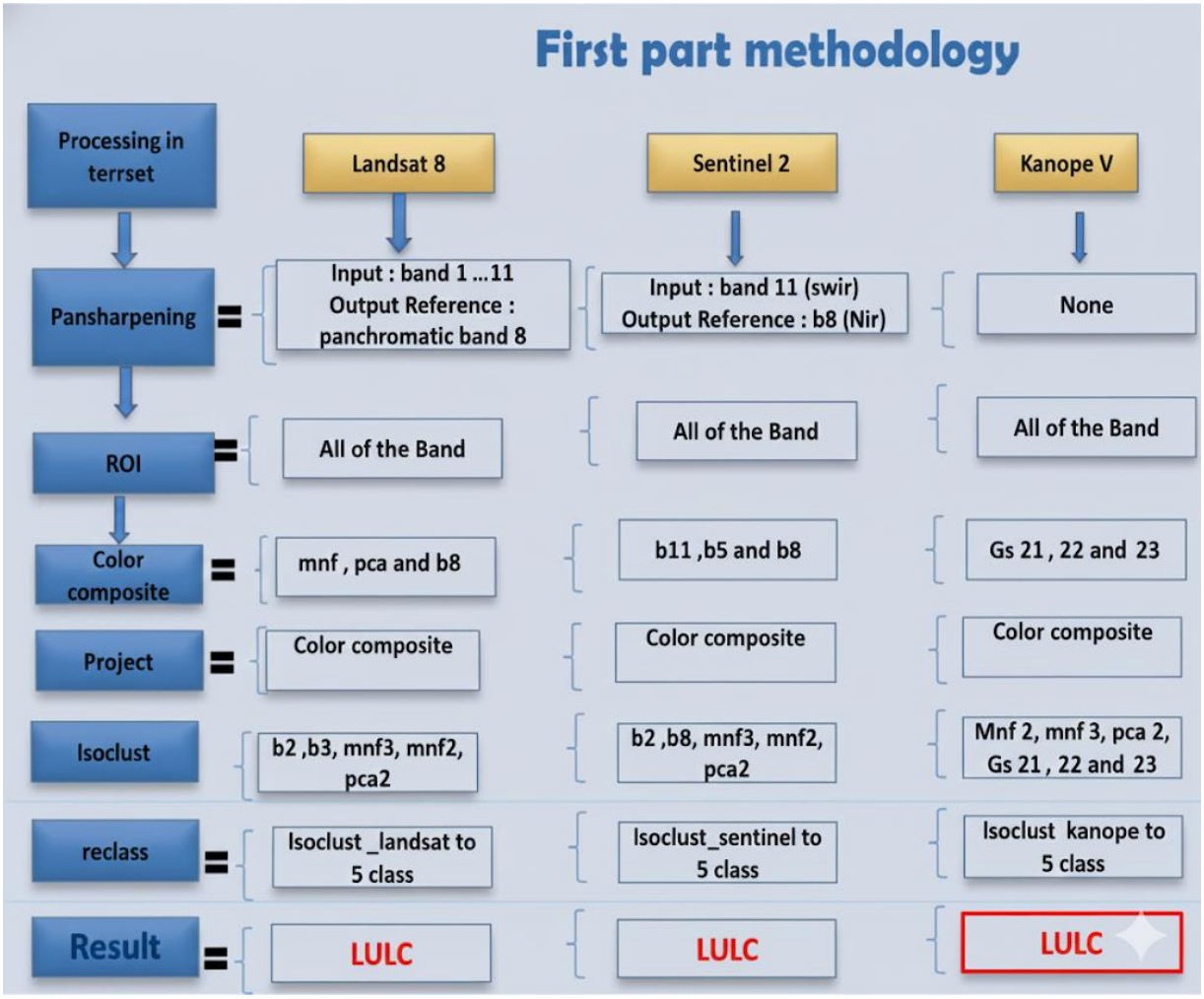
3. Region of Interest (ROI) Definition and Feature Extraction

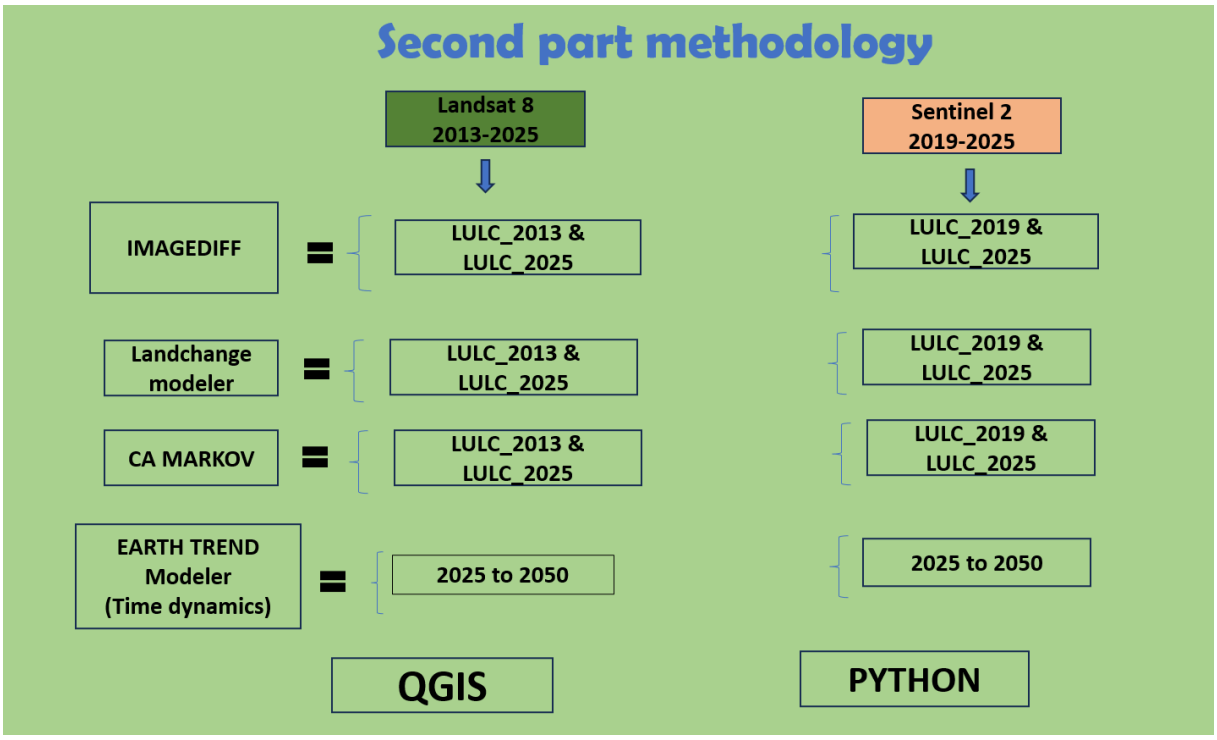
Regions of Interest (ROIs) were defined for all land-cover classes using the full spectral information available for each sensor. For Landsat 8/9, ROIs were built using all available bands, with particular emphasis on Band 8 in combination with color composites, Minimum Noise Fraction (MNF), and Principal Component Analysis (PCA) to improve class separability. For Sentinel-2, ROIs were generated using all spectral bands, with enhanced discrimination achieved through combinations of Band 11 (SWIR), Band 5 (red-edge), and Band 8 (NIR). For Kanopé V imagery, ROIs were constructed using the GS 21, GS 22, and GS 23 bands, which provided detailed spatial and spectral information for urban structures and surface materials. This approach ensured consistent and representative training samples across sensors.

4. Classification Approach

Land-use and land-cover mapping was performed by combining unsupervised and supervised classification approaches within a machine learning framework. Unsupervised classification

was first applied to explore spectral groupings and identify dominant land-cover patterns. Supervised classification was then conducted using machine learning algorithms trained on the previously defined ROIs, allowing the integration of spectral, spatial, and transformed features derived from MNF and PCA. This combined approach improved the robustness of class separation for built-up areas, vegetation, water bodies, soil, and mixed surfaces, while accounting for the differences in spatial resolution and spectral richness between Landsat, Sentinel-2, and Kanopé V data.

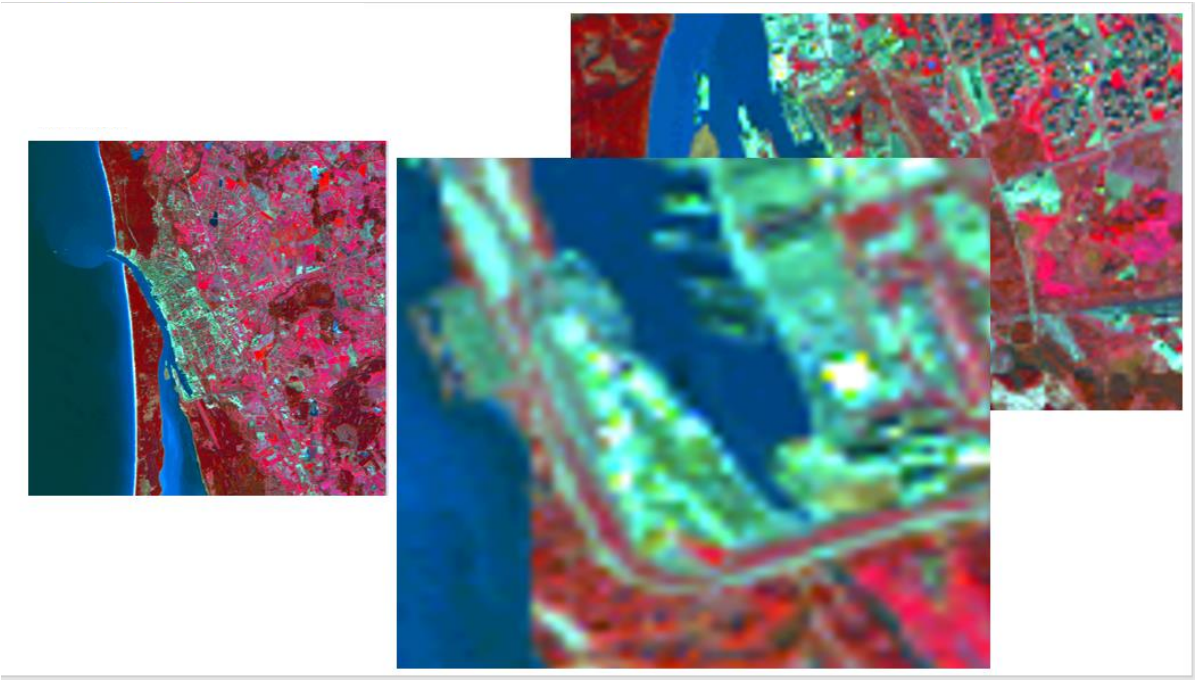



Second part methodology
Landsat 8
2013-2025
Sentinel 2
2019-2025
IMAGEDIFF
LULC_2013 &
LULC_2025
LULC_2019 &
LULC_2025
Landchange
modeler
LULC_2013 &
LULC_2025
LULC_2019 &
LULC_2025
CA MARKOV
LULC_2013 &
LULC_2025
LULC_2019 &
LULC_2025
EARTH TREND
Modeler
(Time dynamics)
2025 to 2050
2025 to 2050
QGIS
PYTHON

PART I:

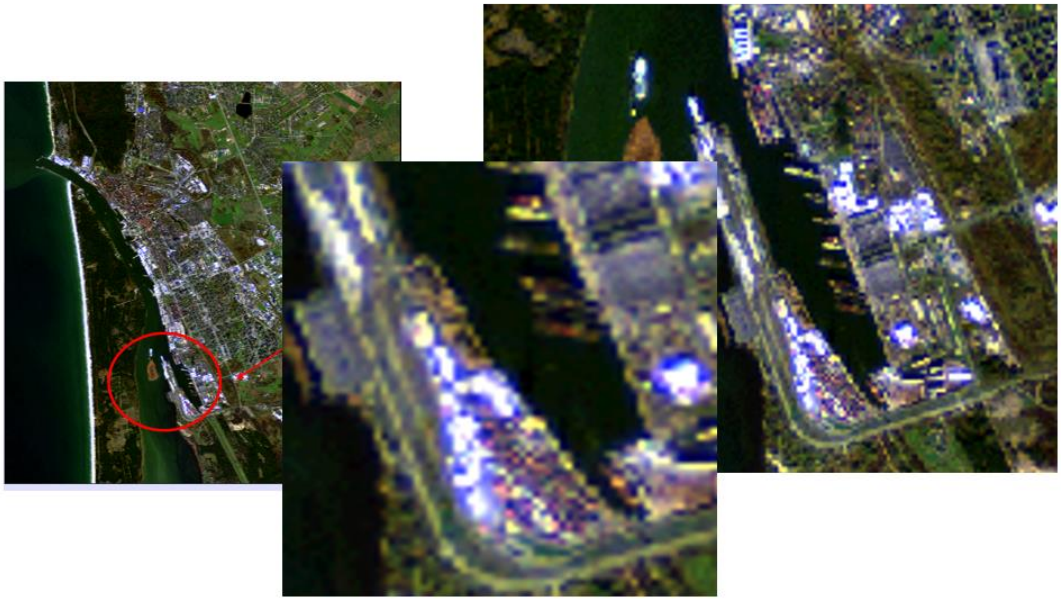
DETECTION, RECOGNITION AND IDENTIFICATION OF URBAN OBJECTS AND STRUCTURES

Figure 1: LANDSAT 8 Preprocessing result



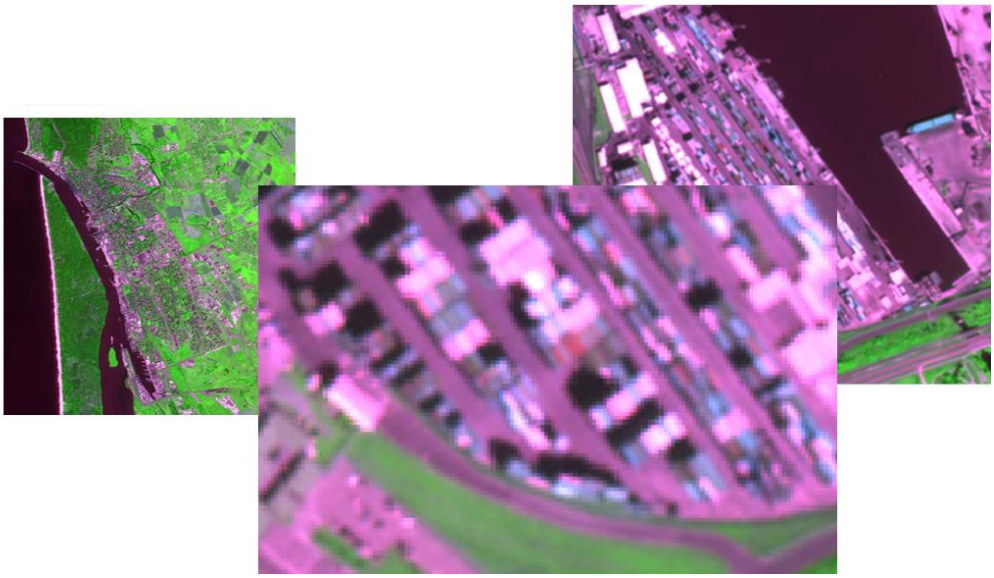
For Landsat 8, all spectral bands from Band 1 to Band 11 were pansharpened using the panchromatic Band 8 as the spatial reference. Color composites were generated in TerrSet by combining Principal Component Analysis (PCA), Minimum Noise Fraction (MNF), and Band 8 in order to enhance urban feature discrimination. Regions of Interest (ROIs) were defined using the full set of pansharpened spectral bands to ensure robust training and classification.

Figure 2: SENTINEL 2 Preprocessing result



For Sentinel-2, the SWIR Band 11 was resampled and spatially enhanced using Band 8 as the reference band to ensure consistent spatial resolution across the dataset. Image processing was performed in TerrSet, and color composites were generated using the combination of Band 11 (SWIR), Band 5 (red-edge), and Band 8 (NIR) to enhance the discrimination of urban, vegetation, and soil features. Regions of Interest (ROIs) were defined using all available spectral bands to support supervised and machine-learning-based classification.

Figure 3: KANOPE V : Preprocessing result



For Kanope V imagery, no pansharpening was applied due to its very high native spatial resolution. Image processing was carried out in TerrSet, and color composites were generated by combining the GS 21, GS 22, and GS 23 bands to enhance the visualization and discrimination of urban structures and surface materials. Regions of Interest (ROIs) were defined using all available spectral bands to support detailed object-level analysis and classification.

5. Temporal Analysis

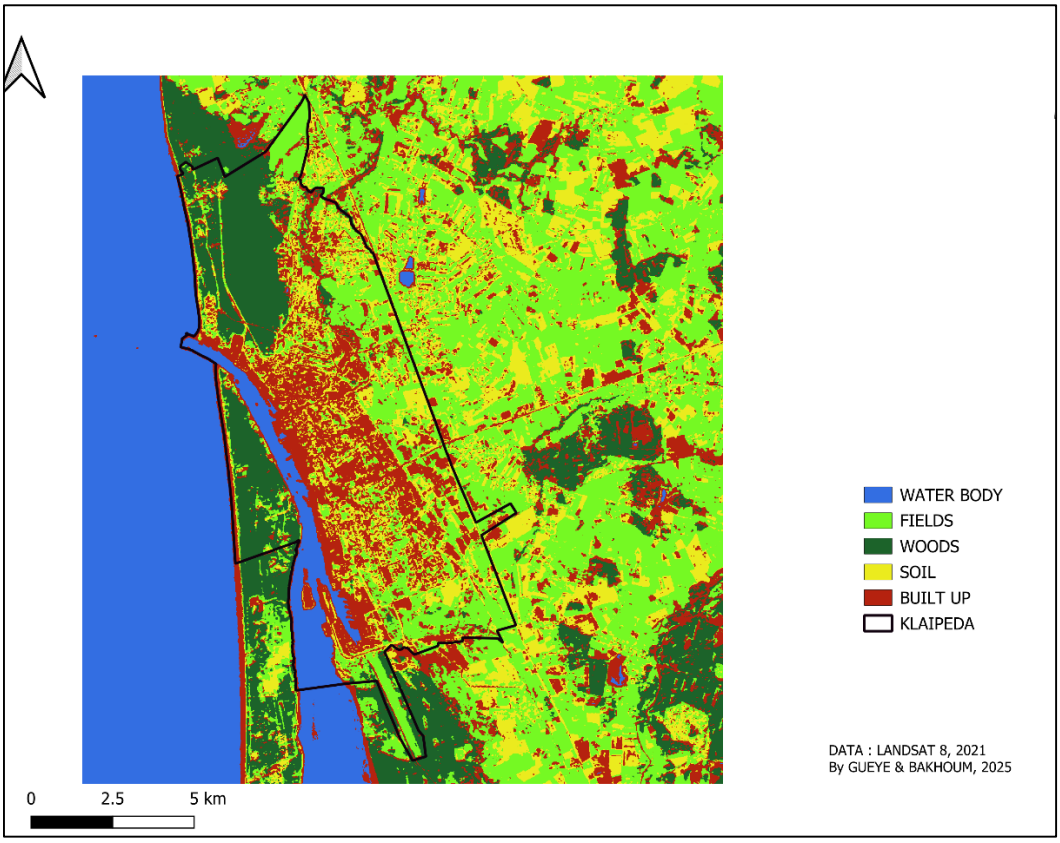
The classification methodology was applied consistently to multi-temporal Landsat and Sentinel-2 imagery from 2013 to 2025 using the same seasonal periods. Classified maps were compared through post-classification analysis to identify spatial and temporal changes in urban expansion, coastal dynamics, and land-cover transitions. The use of harmonized processing steps and machine learning-based classification enabled a consistent comparison of urban and coastal transformations across sensors and time.

1. Comparisons of sensors

1.1. Landsat sensors

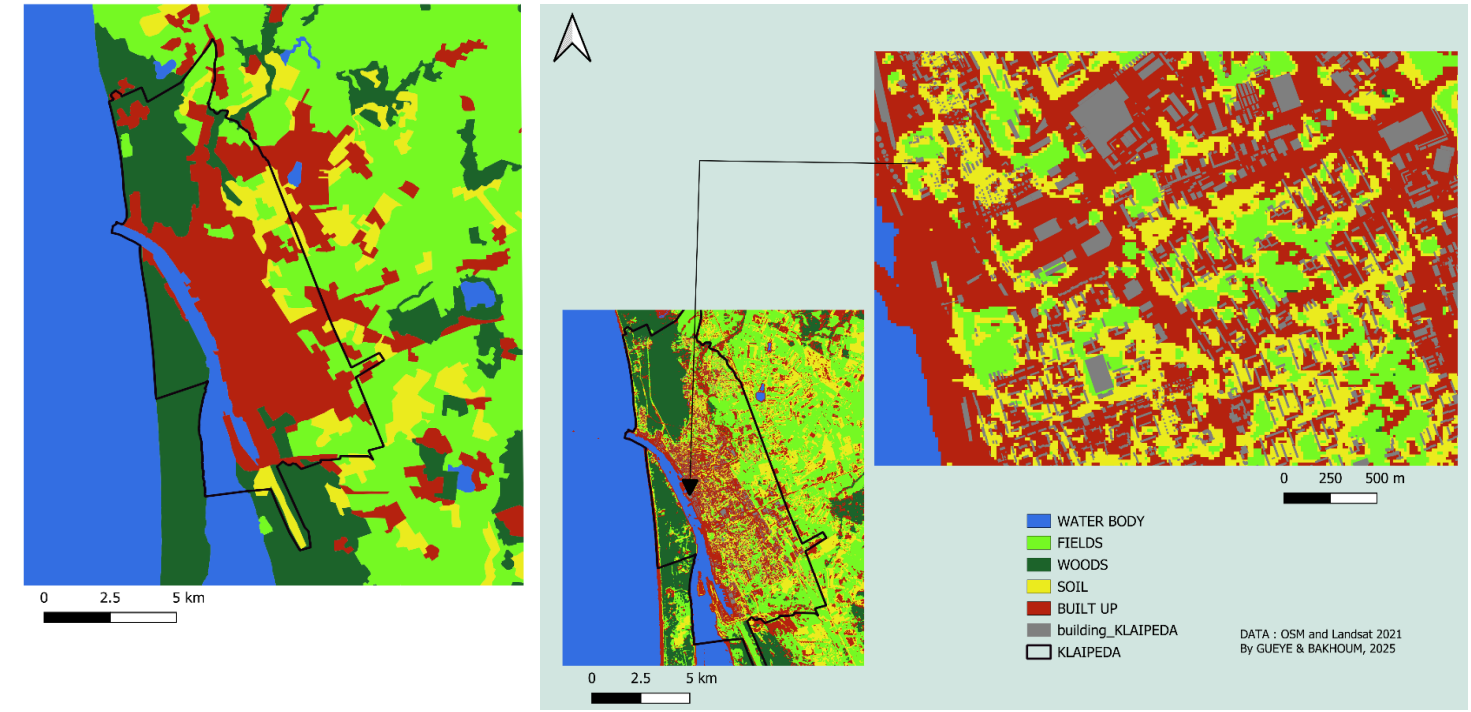
1.1.1. Landsat 8 classification results

Figure 4: Land use land cover Klaipeda



The land use and land cover (LULC) map of the Klaipėda region was produced using Landsat 8 imagery following a dedicated classification methodology and a sequence of preprocessing and analytical steps. The resulting classification distinguishes five major land-cover classes: water bodies, agricultural fields, wooded areas, bare soil, and built-up surfaces. These results provide a basis for subsequent comparison with external reference datasets in order to quantitatively evaluate the detectability and representation of urban objects using Landsat 8 data.

Figure 5: Landsat 8 comparative analysis



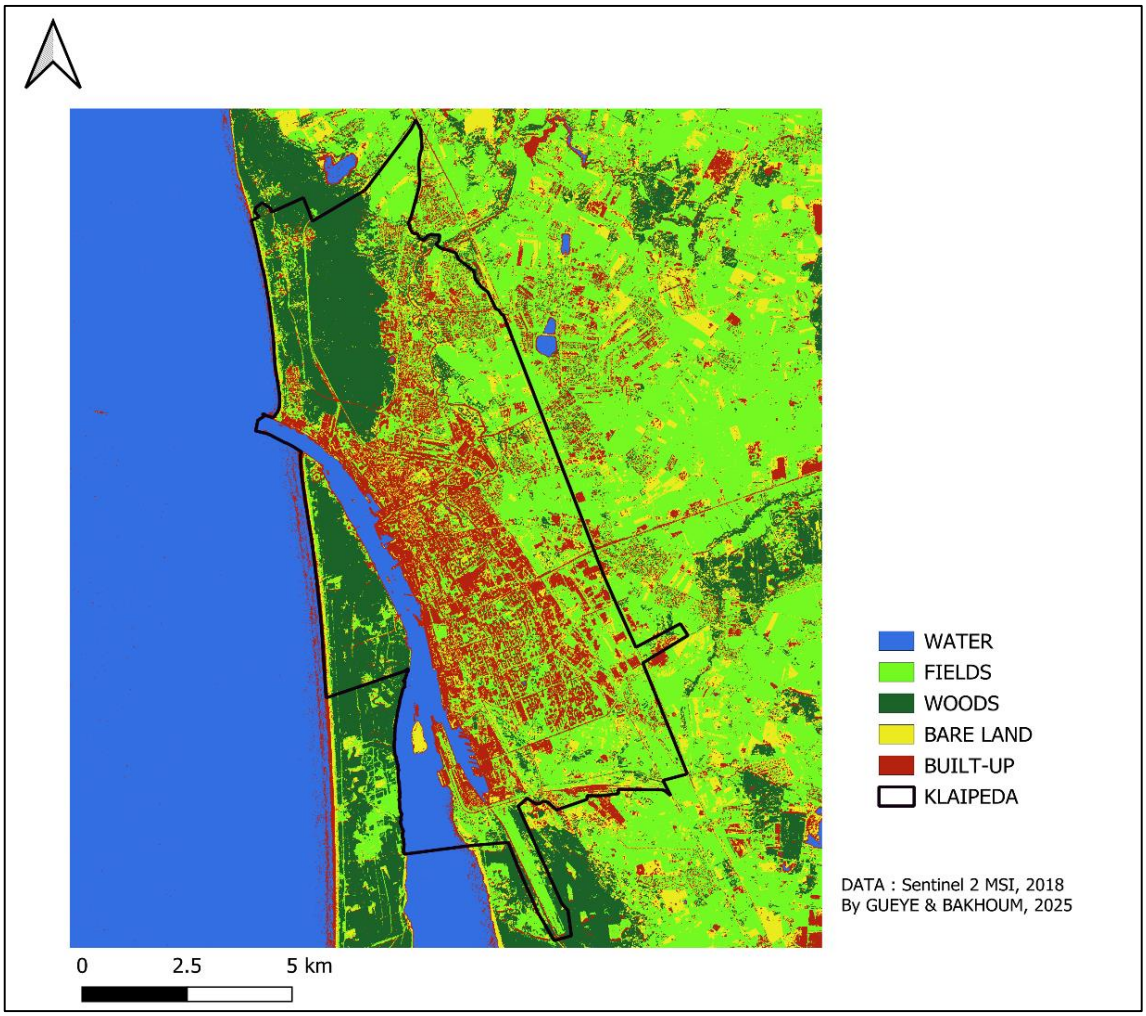
The classification of land-cover objects into five distinct groups makes it possible to identify urban and non-urban elements at a global scale and to analyze the overall spatial structure of the study area. To better evaluate the reliability of the classification derived from Landsat 8 data, land-cover classes sharing identical codes in the CORINE Land Cover (CLC) database were aggregated in order to generate a simplified five-class CLC map comparable to the LULC classification. This harmonization enabled a consistent comparison between the two datasets.

At a general level, the spatial distribution of land-cover classes shows a strong similarity between the Landsat-based LULC map and the CLC map, particularly for major classes such as water bodies, wooded areas, and agricultural fields. However, notable discrepancies can be observed for certain classes, especially the built-up class in the LULC results. This class appears to include some areas that correspond to agricultural fields in the CLC dataset, which represents an abnormal classification behavior. Such misclassification is likely related to spectral confusion between built-up surfaces and certain agricultural or bare soil areas, as well as to similarities in their spectral responses at the spatial resolution of Landsat 8 imagery.

1.2.Sentinel sensor

1.2.1. Sentinel 2 classification results

Figure 6: Sentinel 2 classification result

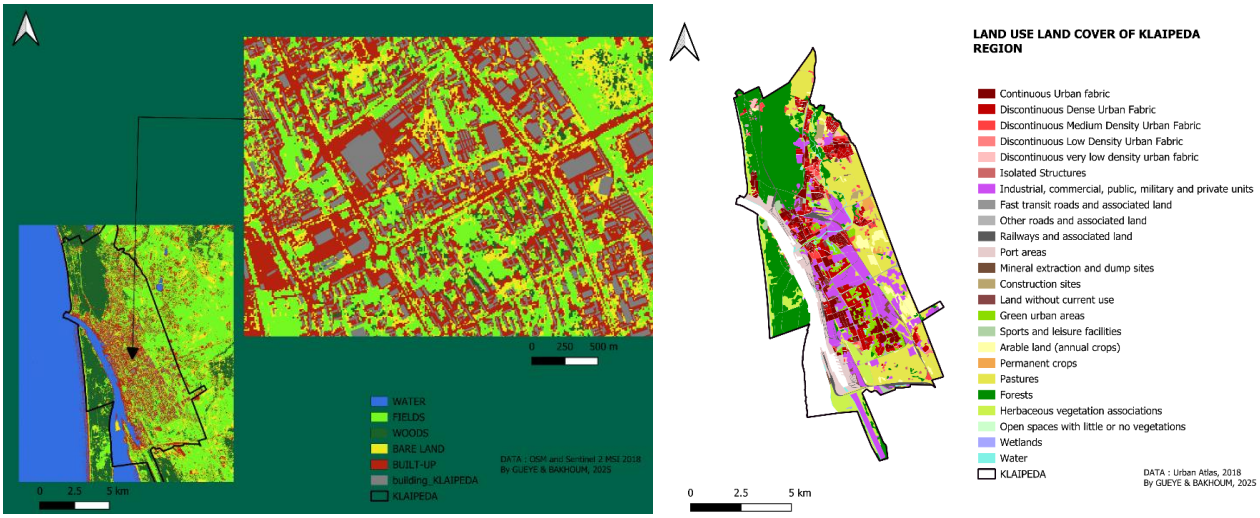


This land use and land cover (LULC) map of the Klaipėda region was generated using Sentinel-2 imagery, following a structured methodology and a series of preprocessing and analytical steps. The classification identifies five primary land-cover classes: water bodies, agricultural fields, wooded areas, bare soil, and built-up surfaces.

Thanks to Sentinel-2's higher spatial resolution (10–20 m for most spectral bands) and its extended spectral coverage, including visible, near-infrared, and shortwave infrared bands, this dataset enables a more detailed representation of land-cover patterns compared to coarser-resolution sensors. The resulting LULC map provides a basis for future comparisons with reference datasets such as CORINE Land Cover (CLC), allowing for the detectability and spatial representation of urban objects using Sentinel-2 imagery.

1.2.2. Sentinel 2 comparative analysis

Figure 7: Sentinel 2 comparative analysis

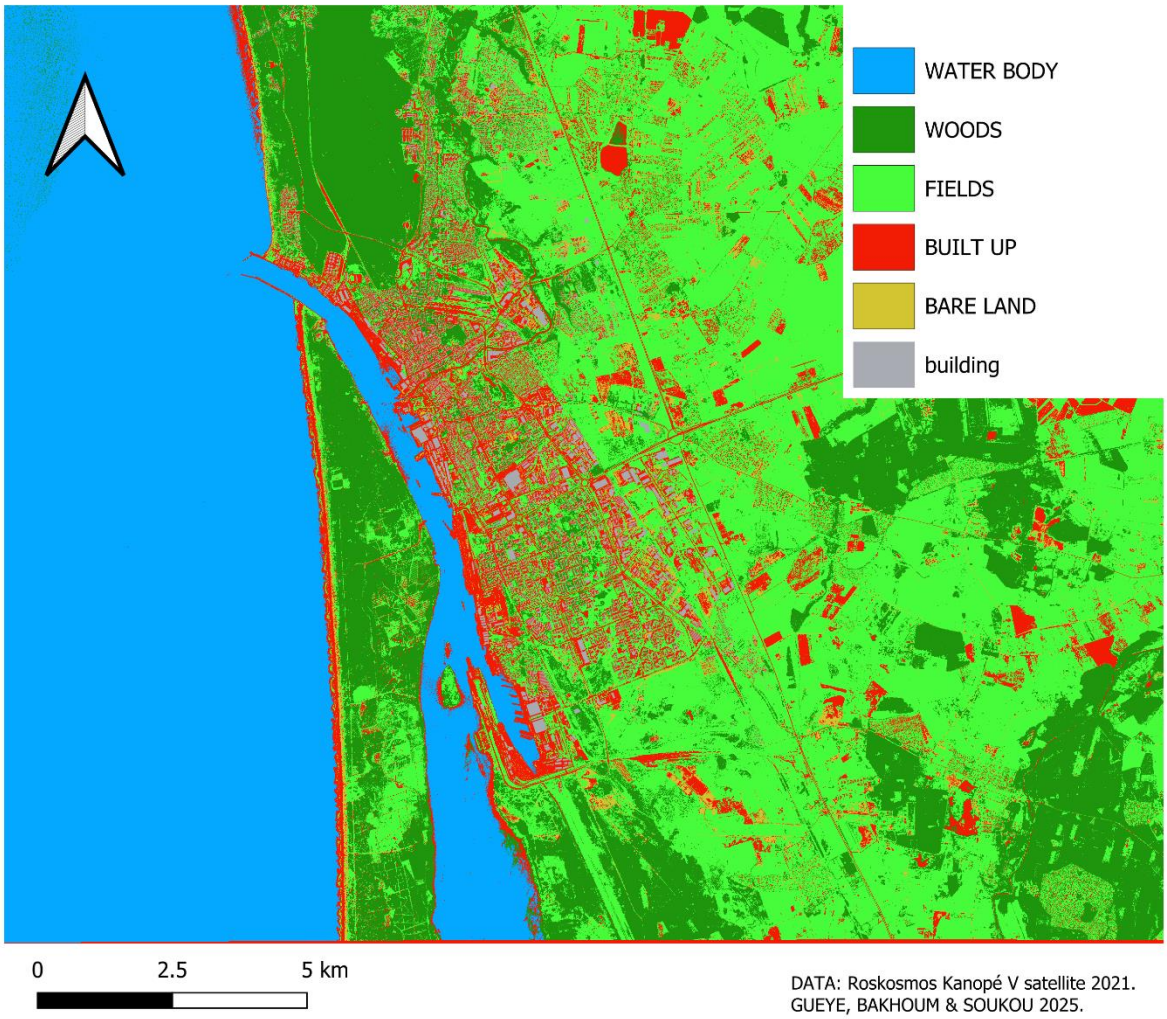


The LULC map of the Klaipéda region derived from Sentinel-2 imagery enables a detailed assessment of urban and non-urban land-cover patterns and can be directly compared with CORINE Land Cover (CLC) data. Sentinel-2's higher spatial resolution (10–20 m) and extended spectral coverage, including visible, near-infrared, and shortwave infrared bands, allow for a more precise delineation of built-up areas, agricultural fields, forests, water bodies, and bare soil compared to the coarser classes represented in CLC. Overall, there is a strong correspondence between the Sentinel-2 classification and the CLC map for major land-cover categories, such as water bodies, wooded areas, and agricultural fields. However, discrepancies are observed in urban classes: some dense built-up zones in Sentinel-2 are mapped as less dense or mixed classes in CLC, while smaller urban structures and narrow features are often underrepresented in CLC due to its minimum mapping unit. These differences highlight the advantages of Sentinel-2 in capturing finer spatial details and improving the detectability of urban objects, while also revealing limitations in existing reference datasets for high-resolution urban monitoring.

1.3. Kanope sensor

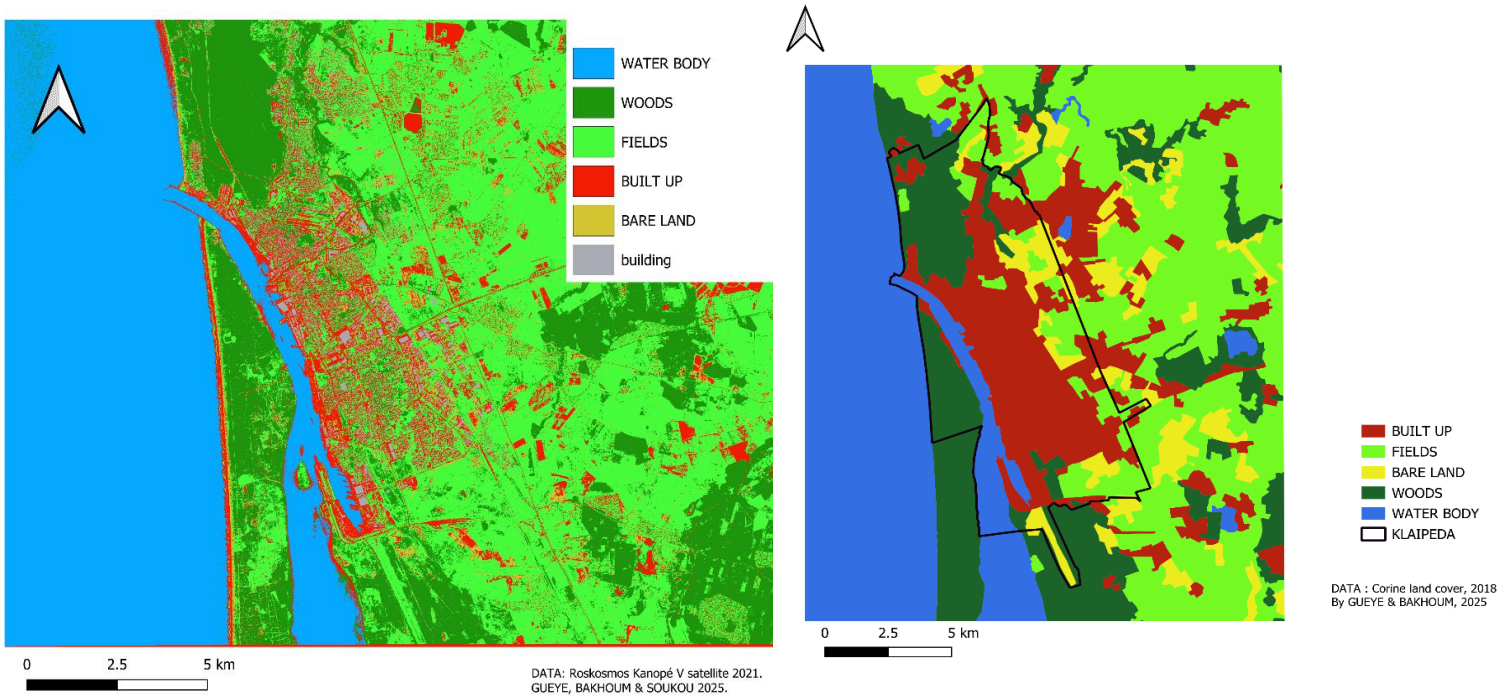
1.3.1. Kanope V classification results

Figure 8: Kanope V classification results



1.3.2 Kanope V classification results

Figure 9: Kanope V classification results



This land use and land cover (LULC) map of the Klaipėda region was produced using Kanope V imagery, following a dedicated methodology and a series of processing steps. The classification distinguishes five major land-cover classes: water bodies, fields, wooded areas, bare soil, and built-up surfaces.

Based on these results, comparisons with selected reference datasets will be conducted to more precisely evaluate the detectability and accurate representation of urban objects using Kanopé V imagery. Given its very high spatial resolution, Kanopé V allows for a more detailed identification of urban structures and finer-scale land-cover features compared to medium-resolution sensors such as Landsat 8.

1.1.1. Kanope V vs Urban Atlas

Figure 11: Klapeida LULC, Urban Atlas

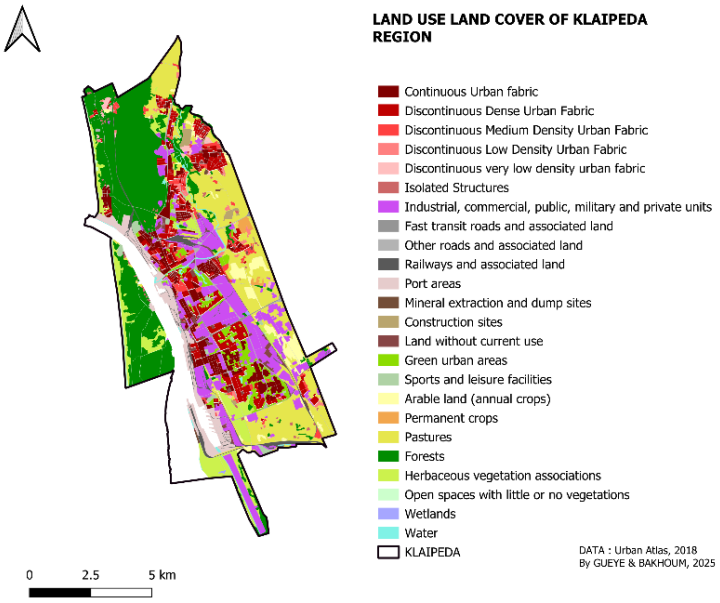
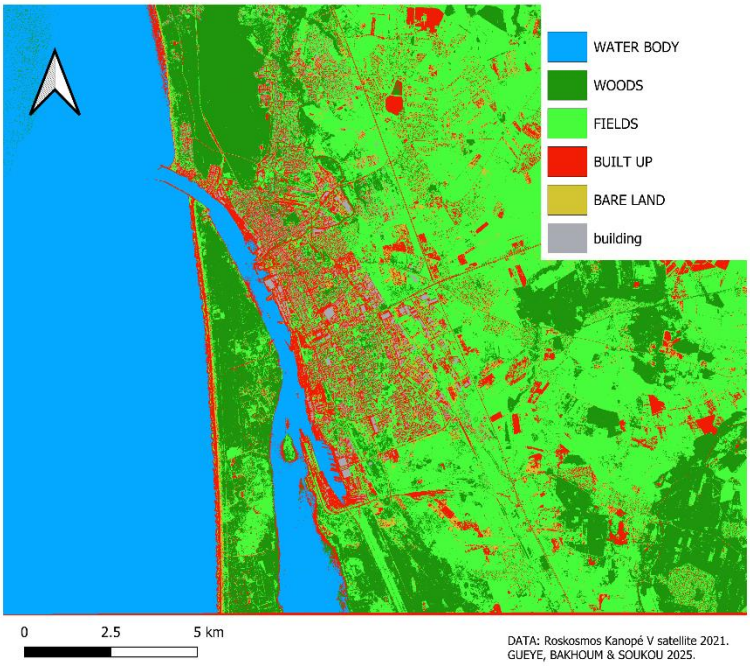


Figure 10: LULC Kanopé V



The Urban Atlas and OpenStreetMap (OSM) building footprint data provide additional precision for evaluating the Landsat 8 LULC results. The Urban Atlas dataset offers detailed information on the structure of each land-cover type, including dense and less dense urban areas, commercial zones, and industrial areas, which allows for a clearer identification of specific urban objects. Similarly, OSM data provide accurate building outlines, enabling a comparison with the Landsat 8 LULC map to assess the classification accuracy and the detectability level of urban structures.

However, when examining finer spatial scales, it becomes apparent that some constructions are not correctly aligned with the built-up areas in the LULC map, and certain building shapes are not accurately represented. These discrepancies are likely due to the spatial resolution of Landsat 8 (15 m after resampling), which may be insufficient to fully capture the geometry and precise location of smaller urban objects. Consequently, while Landsat 8 provides a reliable overview of urban land cover, its limitations in spatial detail can affect the accurate identification of individual buildings and complex urban structures.

1.2. Part one conclusion

Tableau 1: The comparison table of Landsat 8, Sentinel-2, and Kanopé V

Object/detectability level	Landsat 8	Sentinel 2	kanope
Built up	High	High	High
Fields	Rather low	Medium	High
Woods	Low	Medium	High
Water	Too low	Low	High
Bare land	Very low	Low	Medium

The comparison table of Landsat 8, Sentinel-2, and Kanopé V illustrates the relative strengths and limitations of each sensor in detecting and representing the five primary land-cover classes: water bodies, fields, woods, soil/bare land, and built-up areas. Across all sensors, water bodies are consistently well captured due to their distinct spectral signature, with Kanopé V providing the most precise delineation of narrow rivers and small ponds. Agricultural fields are clearly identified in Sentinel-2 and Kanopé V, while Landsat 8 sometimes shows confusion between fields and bare soil, likely due to its coarser spatial resolution. Wooded areas are generally well represented by all three sensors; however, Kanopé V can capture finer-scale forest patches and fragmented vegetation that remain undetected by Landsat 8 and, to a lesser extent, Sentinel-2. Soil and bare land show moderate detectability across sensors, with Sentinel-2 and Kanopé V providing improved spatial detail and spectral discrimination compared to Landsat 8.

The built-up class shows the greatest variability between sensors. Landsat 8 often merges smaller built-up structures with adjacent fields or bare soil, leading to overgeneralization. Sentinel-2 provides better urban detail and can distinguish denser versus less dense urban areas, but small individual buildings may still be unresolved. Kanopé V excels in detecting individual structures, streets, and complex urban patterns, providing the highest spatial fidelity and enabling precise urban object mapping. Overall, the table demonstrates a trade-off between coverage, spectral resolution, and spatial detail, highlighting that higher-resolution sensors like Kanopé V are essential for detailed urban studies, while Sentinel-2 offers a practical balance for medium-scale monitoring, and Landsat 8 provides long-term temporal consistency.

PART II:

TEMPORAL ANALYSIS OF RECENT CHANGES IN KLAIPĖDA

2. Temporal analysis of Klapeida

2.1. Satellite Data Overview

This study is based on multi-temporal optical satellite imagery from Sentinel-2 MSI and Landsat 8/9 OLI–TIRS to analyze land cover and land use changes in the Klaipėda region and its coastline. To ensure temporal consistency and reduce seasonal effects, imagery was selected from the same period of the year (September) for all datasets.

2.1.2. Sentinel-2 MSI Data

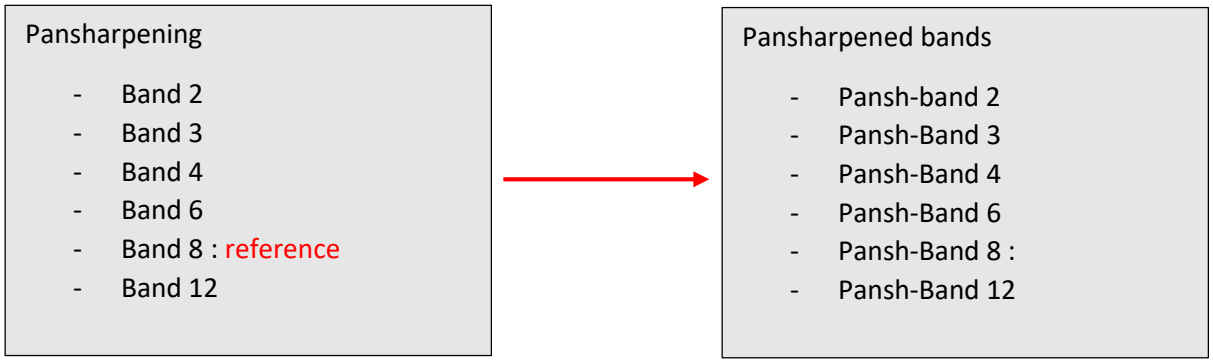
Sentinel 2 MultiSpectral Instrument (MSI) imagery was selected for two reference dates representing recent conditions:

- **September 2019**
- **September 2025**

September was chosen to ensure comparable phenological conditions, relatively low cloud cover, and stable surface reflectance. These dates allow the analysis of land cover and land use changes over a 6 year period. This satellite has main characteristics:

- Spatial resolution: 10 m, 20 m, 60 m
- Spectral bands: Visible, NIR, and SWIR
- Revisit time: 5 days

All Sentinel-2 bands used in this study were resampled to 10 m for spatial consistency.



2.1.3. Landsat 8/9 OLI–TIRS Data

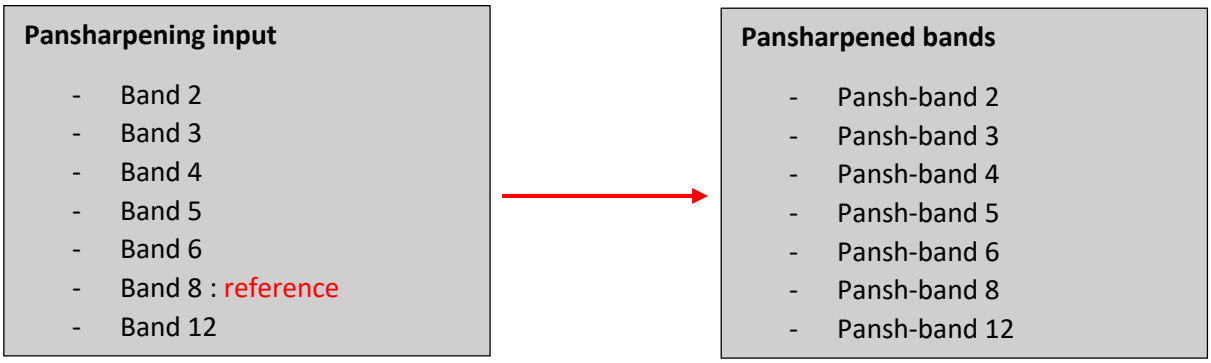
Landsat imagery from the Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) onboard Landsat 8/9 was selected for two dates in order to enable long-term change detection and comparison with Sentinel-2 results:

- **September 2013**
- **September 2025**

These dates allow the analysis of land cover and land use changes over a 12-year period, while maintaining seasonal consistency with Sentinel-2 data. These are the main characteristcs of the satellite:

- OLI multispectral bands: **30 m**
- Panchromatic band: **15 m**
- TIRS thermal bands: **100 m (resampled to 30 m)**

Landsat imagery was **pan-sharpened and resampled to 15 m** to improve spatial detail and facilitate comparison with Sentinel-2 data.



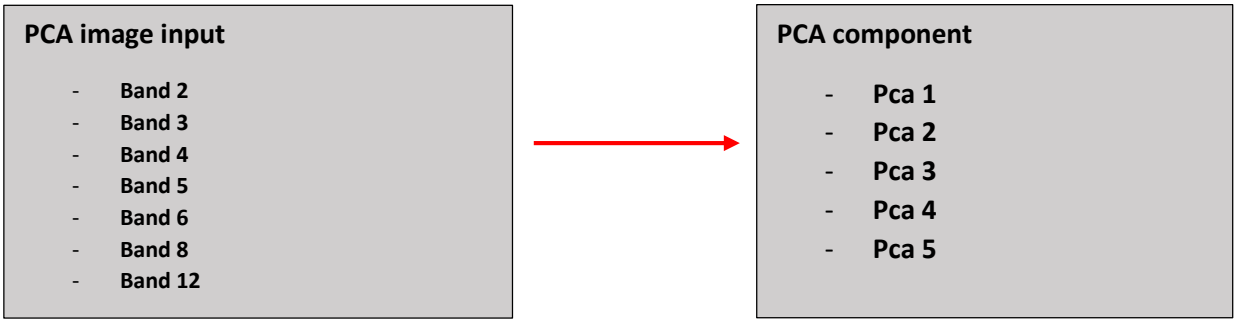
2.2. Methodological Flowchart

2.2.1. Data Pre-processing

In addition to the pan-sharpening step described previously, several further pre-processing and feature extraction techniques were applied prior to the spatio-temporal analysis. These included Principal Component Analysis (PCA), Minimum Noise Fraction (MNF) transformation, the computation of spectral indices, and the definition of Regions of Interest (ROIs).

• Principal Component Analysis (PCA):

PCA was used to reduce data dimensionality and to concentrate the maximum spectral variance into a smaller number of uncorrelated components, facilitating change detection and interpretation.



- **Minimum Noise Fraction (MNF):**

The MNF transformation was applied to separate noise from useful spectral information and to enhance signal quality, thereby improving the robustness of subsequent analyses.

MNF image input

- **Band 2**
- **Band 3**
- **Band 4**
- **Band 5**
- **Band 6**
- **Band 8**
- **Band 12**

→

MNF component

- **MNF 1**
- **MNF 2**
- **MNF 3**
- **MNF 4**
- **MNF 5**

- **Spectral Indices:**

Selected spectral indices were calculated to emphasize specific surface characteristics such as vegetation, built-up areas, and water bodies, supporting land cover and land use discrimination.

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

We used the NDVI to assess the condition of the vegetation and to distinguish vegetation from crop fields.

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$$

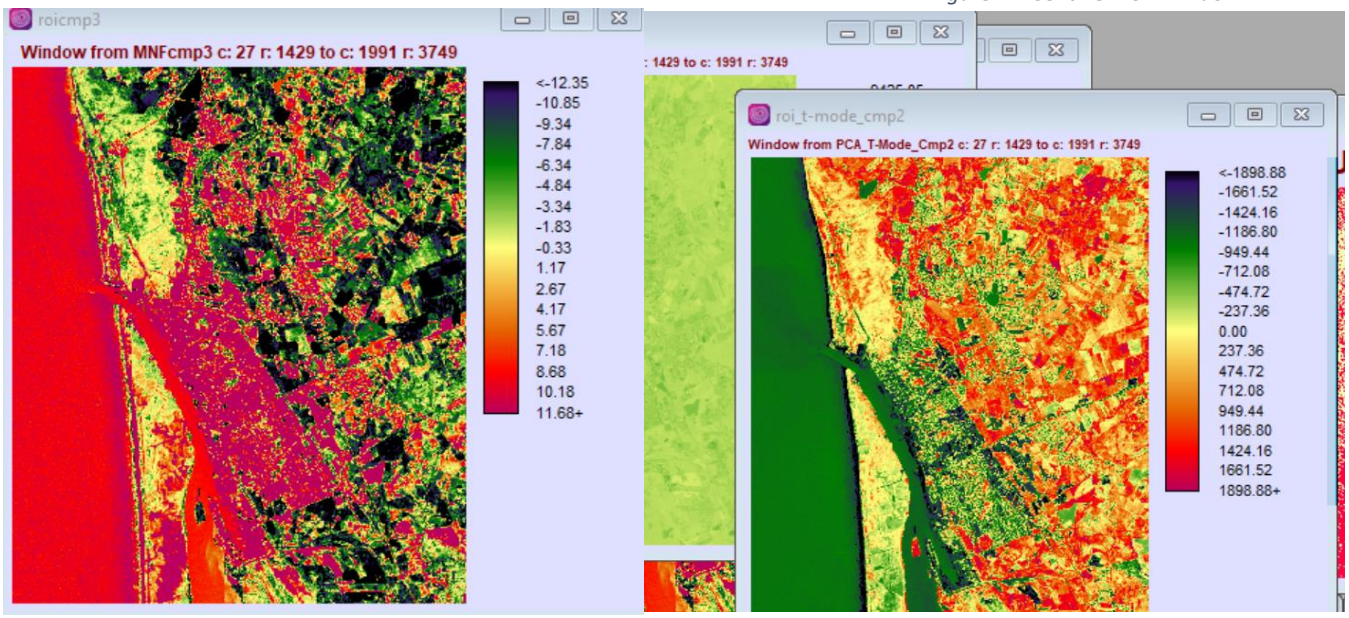
The NDBI was used to detect and separate built-up areas from other land cover types

• **Regions of Interest (ROI):**

ROIs were defined to represent homogeneous land cover classes and were used for training, validation, and statistical analysis.

Figure 13: Sentinel MNF Window

Figure 12: Sentinel PCA Window



2.3. Land Cover and Land Use Change Mapping and Analysis

2.3.1. Sentinel classification Results

Figure 14: Sept 2019 Sentinel 2 LULC

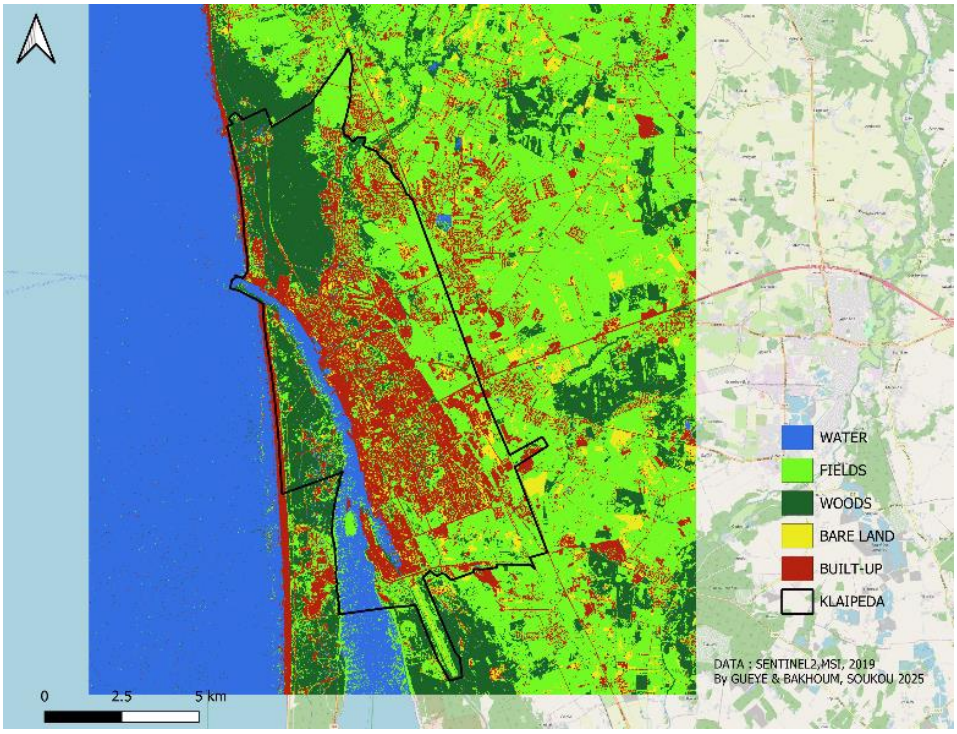
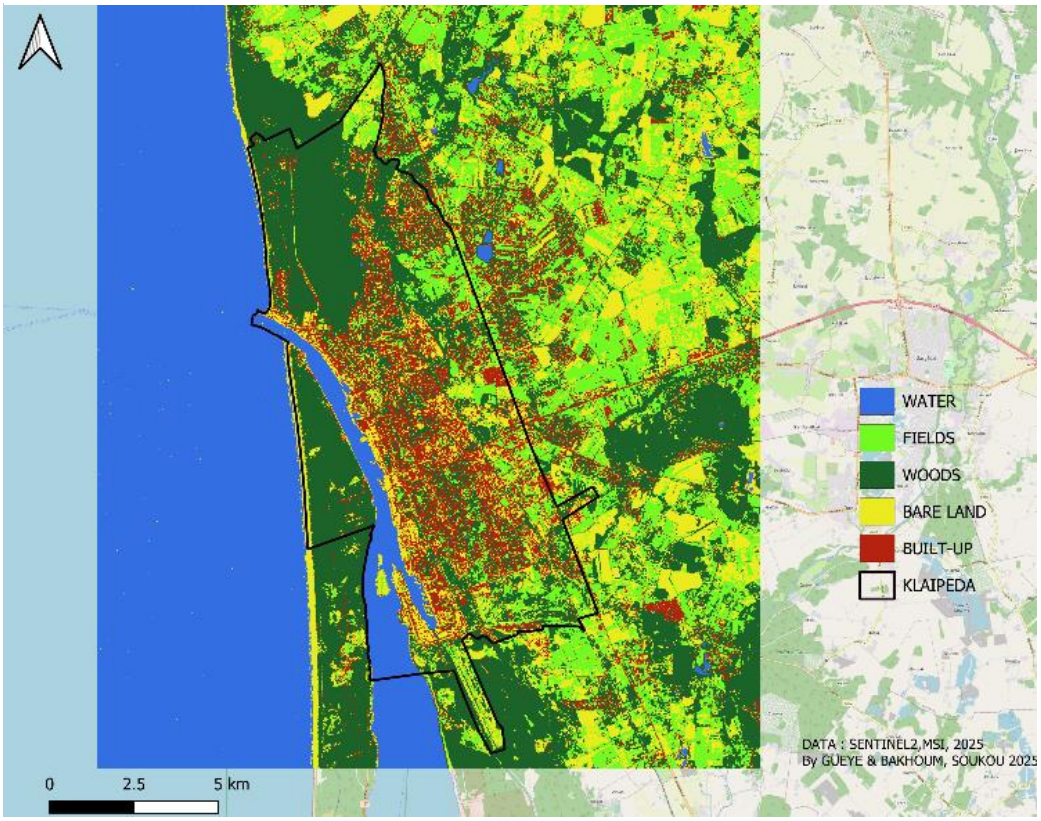


Figure 15: 2025 Sentinel 2 LULC



The Reference Model used for the 2019 and 2025 sentinel 2 classification is the classification processing used for the same sensor in the part A of the project.

The 2025 Sentinel-2 LULC map provides a detailed spatial representation of land use and land cover patterns across the Klaipėda region. Built-up areas (in red) are highly concentrated along the coastal strip and around the urban core, indicating strong urban development pressure. Fields and agricultural areas (light green) dominate the inland landscape, while wooded zones (dark green) form larger continuous patches mainly toward the northern and southern parts of the region. Bare land is more scattered and appears in limited proportions. Water bodies, including coastal and lagoon areas, are clearly distinguishable along the western boundary. Overall, the map reveals a complex land-use mosaic shaped by intensive urbanization near the coast and more rural landscapes inland, reflecting the region's social, environmental, and economic dynamics.

- **Changes Analysis of Sentinel classification:**

Figure 17: Sentinel image diff

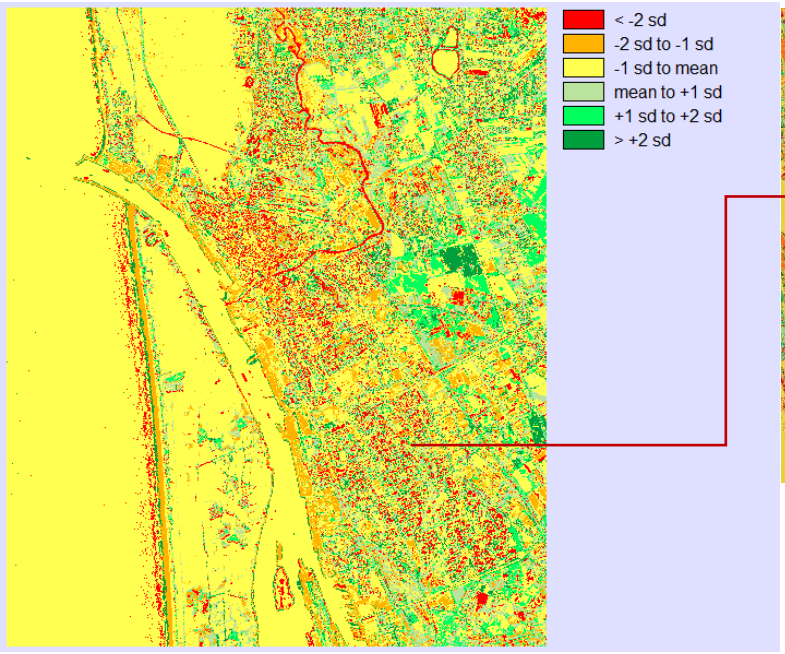
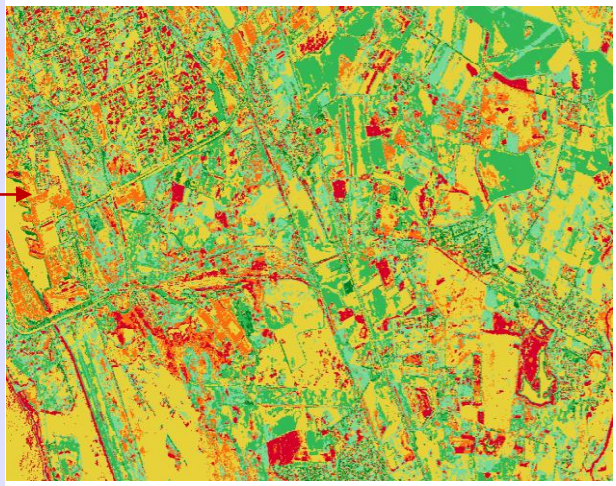
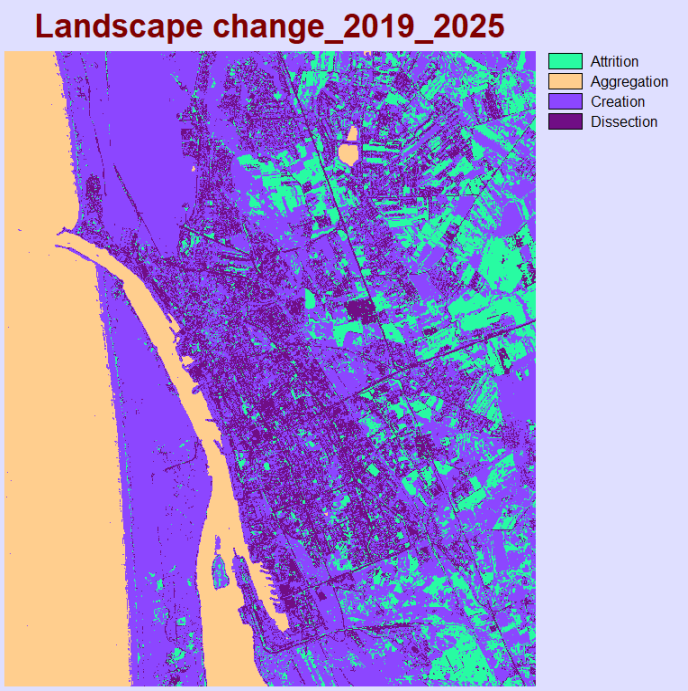


Figure 16: Zoom Sentinel image diff



The map generated from the Image Diff module using the Standardized Class Image option highlights the spatial changes between the two study dates. Most of the pixels fall within the classes ranging from –1 to +1 standard deviation, indicating relative stability of the land cover over the study period. However, some localized areas with values lower than –1 and –2 standard deviations reveal significant negative changes, likely related to urbanization processes, land clearing, or vegetation cover degradation. Conversely, areas showing values higher than +1 and +2 standard deviations indicate marked positive changes, possibly corresponding to vegetation regeneration, expansion of cultivated areas, or reforestation. The heterogeneous distribution of the classes reflects the contrasting dynamics of the studied landscape.

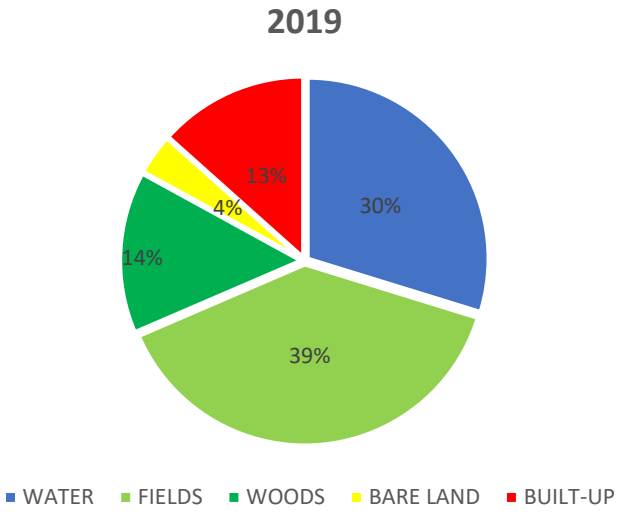
Figure 18:Sentinel Landscape change



This map highlights three major habitat change processes that are essential for understanding biodiversity dynamics between the two study dates. Attrition represents the disappearance of small, isolated habitat patches, indicating localized habitat loss and increased fragmentation that can negatively affect species movement and survival. Aggregation refers to the merging of adjacent habitat patches into larger, more continuous areas, which generally strengthens ecological connectivity and improves habitat quality. Finally, Creation identifies the emergence of new habitat patches in areas previously lacking suitable conditions, reflecting natural regeneration or restoration processes that contribute to biodiversity gain. Together, these processes provide a detailed view of how habitat structure is evolving and help assess the implications for local ecological resilience.

- **Sentinel Descriptive Statistics: Land change modeler**

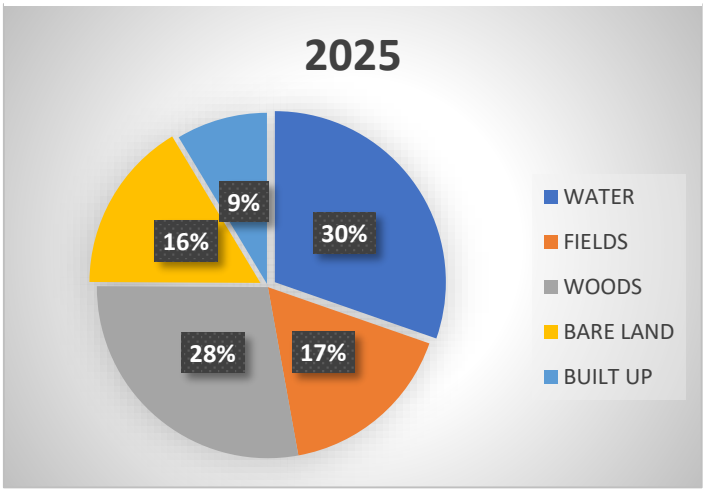
Figure 19: Sentinel Statistics: Land change modeler 2019



The descriptive statistics derived from the 2019 Sentinel 2 data using the Land Change Modeler indicate that agricultural fields represent the dominant land cover class, accounting for 39% of the study area. Water bodies constitute the second largest class with **30%**, reflecting the strong influence of the coastal and aquatic environment in the Klaipėda region. Wooded areas cover 14% of the territory, highlighting the presence of forested and semi-natural landscapes. Built-up areas represent 13%, indicating a significant level of urbanization associated with residential, industrial, and

port-related activities. Finally, bare land occupies a relatively small proportion of the area **(**4%**),** suggesting limited exposed or transitional surfaces at the time of observation. Overall, these results illustrate a landscape dominated by agricultural and aquatic environments, with notable urban and forest components shaping the spatial structure of the region in 2019.

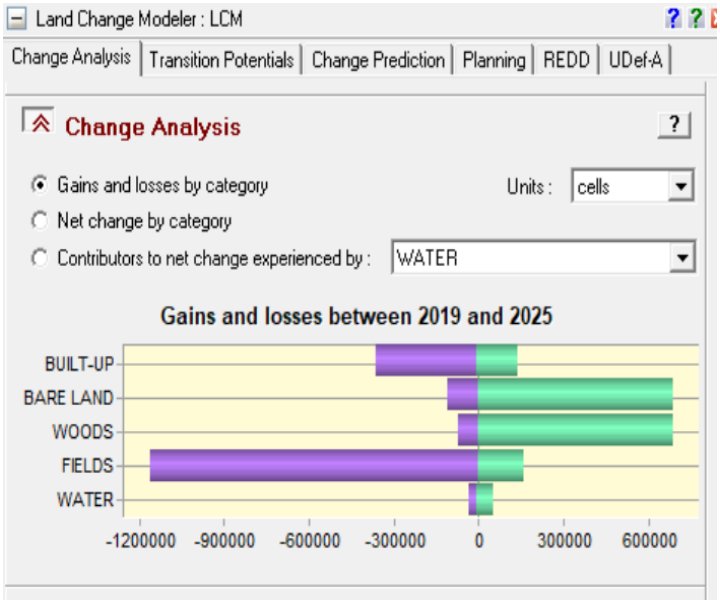
Figure 20: Sentinel Statistics: Land change modeler 2025



The descriptive statistics derived from the 2025 Sentinel 2 data reveal notable changes in land cover distribution within the study area. Water bodies remain the dominant class, accounting for 30% of the area, indicating a relative stability of aquatic and coastal surfaces over the study period. Wooded areas represent 28%**,** showing a substantial increase compared to 2019 and suggesting forest expansion or improved vegetation recovery. Agricultural fields cover 17% of the area, reflecting a marked decline that may be associated with land abandonment, land-use conversion, or reforestation processes. Bare land accounts for 16%, indicating an increase in exposed or transitional surfaces, potentially linked to construction activities, coastal dynamics, or land-use changes. Finally, built-up areas represent 9%, showing a decrease relative to 2019, which may reflect reclassification effects, urban densification, or shifts in land-use patterns. Overall, the 2025 results point to a landscape undergoing significant structural transformation, characterized by reduced agricultural dominance and increased natural or transitional land cover types.

- **Gain and loses by category**

Figure 21: Sentinel 2 LULC Gain and loses by category

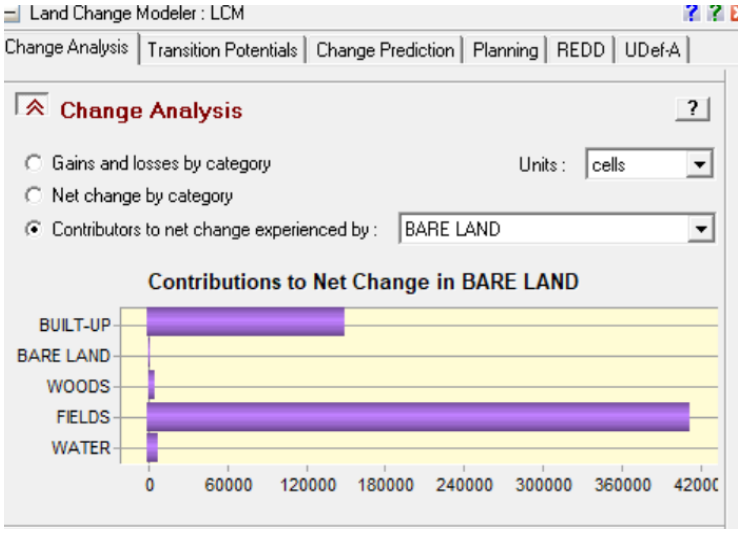


This analysis highlights the gains and losses of each land cover category between 2019 and 2025. For the built up category, losses exceed gains, with more than **300.000** m² of losses compared to approximately **145.000** m² of gains, indicating a net reduction in built-up surfaces. In contrast, bare soil shows a predominance of gains over losses, with nearly **150.000** m² of gains compared to about **60.000** m² of losses, suggesting an expansion of exposed or transitional land surfaces.

The wooded areas also exhibit a net increase, as gains (almost 100.000 m²) exceed losses (approximately 60.000 m²), pointing to forest expansion or vegetation recovery processes. Conversely, agricultural fields experience substantial net losses, with more than 1.100.000 m² of losses compared to around 145.000 m² of gains, reflecting a significant decline in agricultural land. Finally, water bodies show relatively minor changes, with gains (nearly **2.000** m²) slightly exceeding losses (about 10.000 m²), indicating overall stability of aquatic surfaces over the study period.

- **Contribution to net change in bare land :**

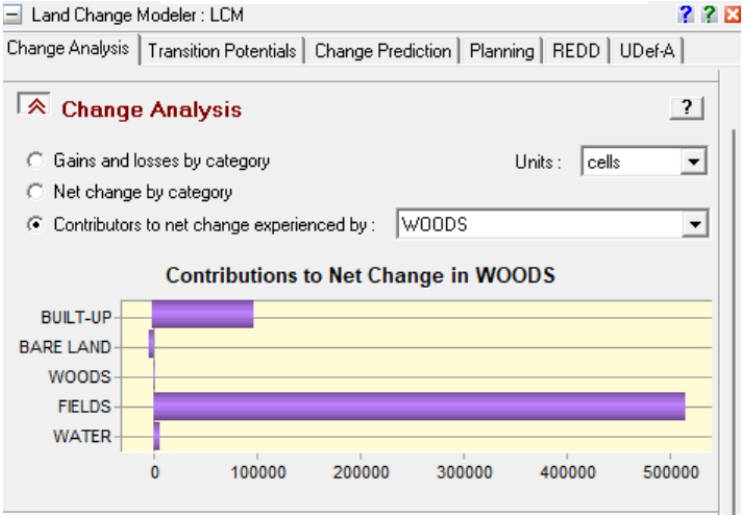
Figure 22: Contribution to net change in bare land (Sentinel 2)



Bare land plays a significant role in the observed land cover transformations. Between 2019 and 2025, more than **120.000** m² of bare land were converted into built-up areas, approximately **1.000** m² transitioned into wooded areas, over **400.000** m² were transformed into agricultural fields, and nearly **2.000** m² were converted into water bodies.

- **Contribution to net change in woods**

Figure 23: Contribution to net change in woods



Wooded areas have also contributed significantly to land cover transitions. Between 2019 and 2025, approximately **90.000** m² of woodland were converted into built-up areas**,** more than **500.000** m² transitioned into agricultural fields, and nearly **1.000** m² were transformed into water bodies.

2.3. Landsat Land Cover and Land Use Change Mapping

2.3.1. Classification Result

Figure 25: 2013 Landsat LULC

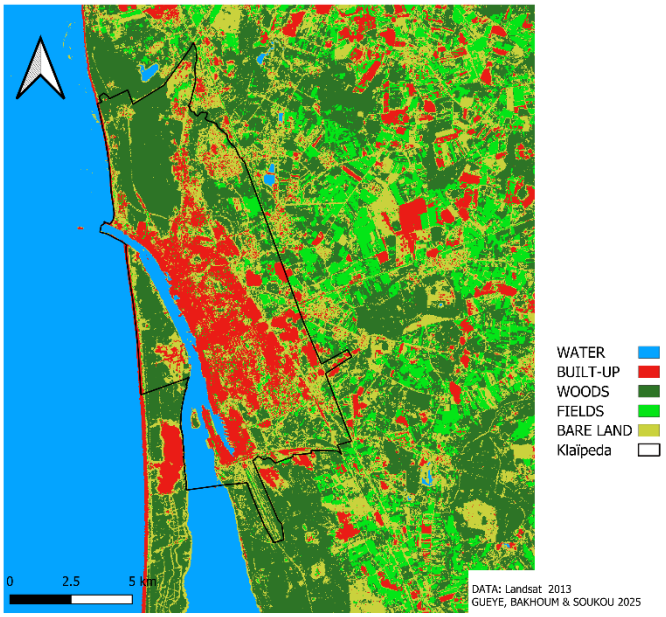
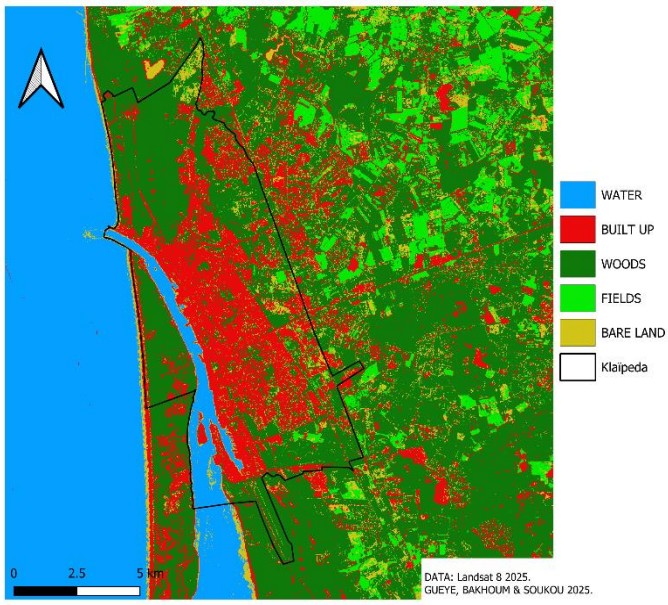


Figure 24: 2025 Landsat LULC



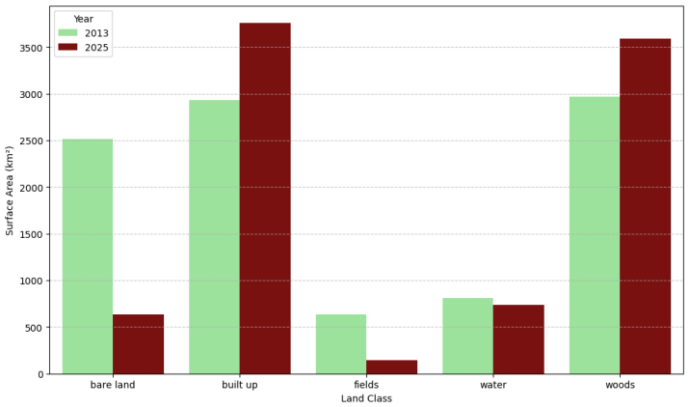
The Reference Model used for the 2013 and 2025 Landsat classification is the classification processing used for the same sensor in the part A of the project.

The 2013 and 2025 LULC maps highlight major landscape changes in the Klaipėda region over the study period. A significant expansion of built-up areas (in red) is visible, particularly around the urban core and along the coastal zone, indicating strong urban growth and spatial densification. At the same time, forested land (dark green) shows a slight reduction, suggesting a potential conversion of woodland into urban or agricultural areas. Field areas (light green) also display spatial changes, reflecting modifications in agricultural use or increasing urban encroachment in rural zones.

Overall, the comparison between 2013 and 2025 reveals increasing anthropogenic pressure, with clear land artificialisation, particularly near coastal areas. These findings underline the importance of sustainable planning strategies to mitigate natural land loss and preserve ecological balance.

Descriptive Statistics: Land change modeler

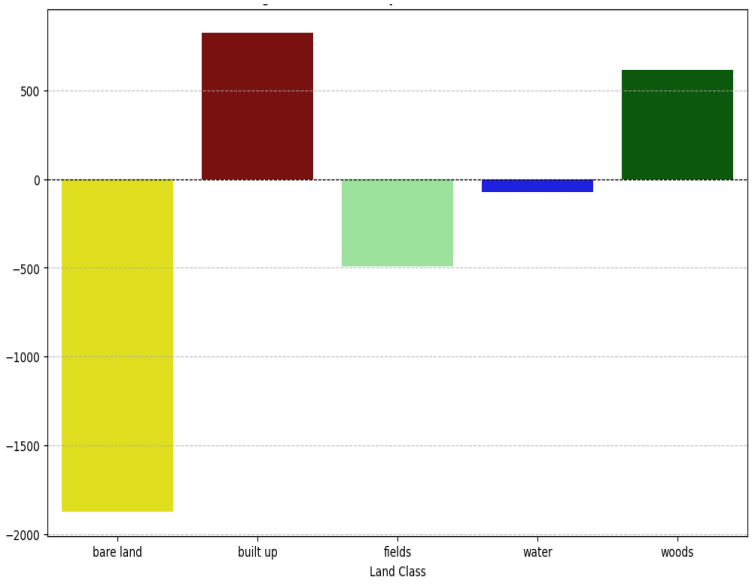
Figure 26: Landsat 2013 - 2025 Land change modeler



The analysis of land surface changes between 2013 and 2025 highlights significant transformations in land use and land cover across the study area. Bare land and agricultural fields experience pronounced declines, indicating extensive conversion to other land uses. In contrast, built-up areas show a substantial increase, reflecting ongoing urban expansion and intensified human activities. Wooded areas also exhibit a notable growth, suggesting processes of reforestation, natural vegetation recovery, or changes in land management practices. Water bodies remain relatively stable throughout the study period, with only minor fluctuations compared to the pronounced changes observed in terrestrial land cover classes.

Gain and loses by category

Figure 27: Gain and loses by category

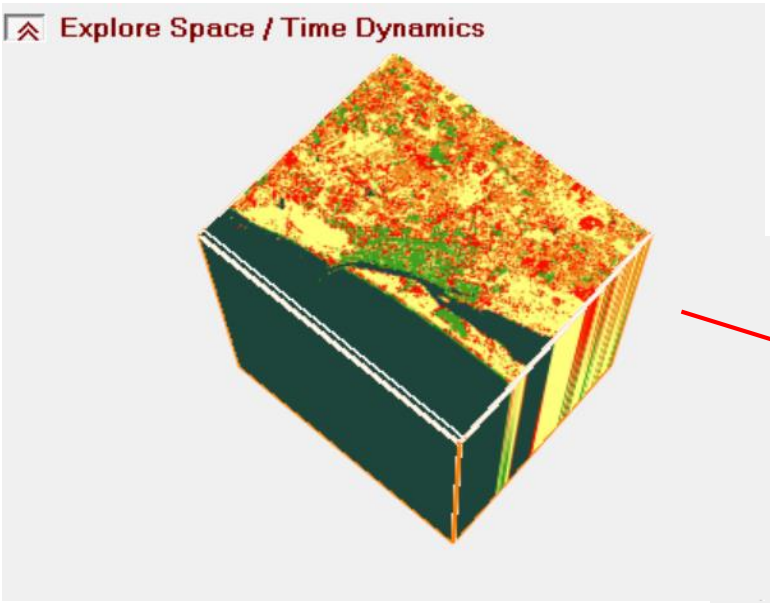


The analysis of land cover changes between 2013 and 2025 reveals clear and contrasting trends among the different land classes. Bare land shows the most significant decline, with a net loss of approximately 1.700 ha**,** indicating widespread conversion of exposed surfaces into other land use types. Agricultural fields also experience a notable reduction of about 498 ha**,** reflecting a substantial decrease in cultivated land during the study period. In contrast, built-up areas undergo marked expansion, with a net gain of approximately 800 ha**,** highlighting continued urban growth and infrastructure development. Wooded areas likewise increase considerably, gaining around 600 ha**,** which may be linked to reforestation processes, natural vegetation recovery, or changes in land management practices. Water bodies

show only a slight decrease of about 50 ha, suggesting a relative stability of aquatic environments compared to the more dynamic changes observed in terrestrial land cover classes.

Perspectives: prediction model

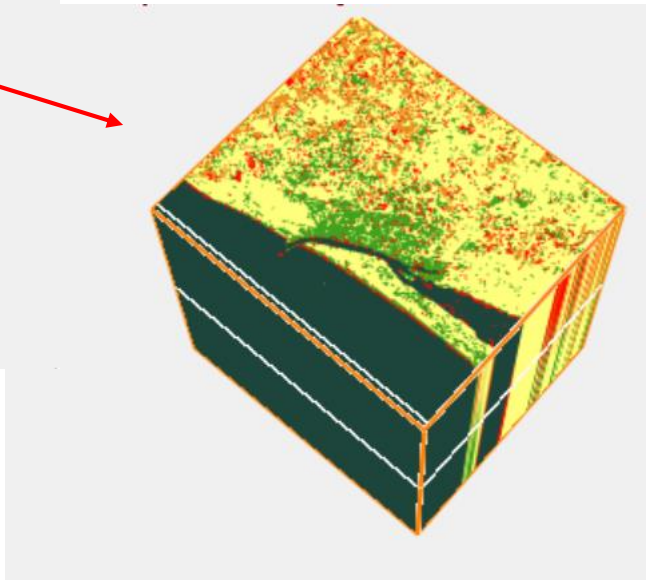
Figure 28: Land use land cover 2025



Land use land cover 2025

Land use land cover 2050

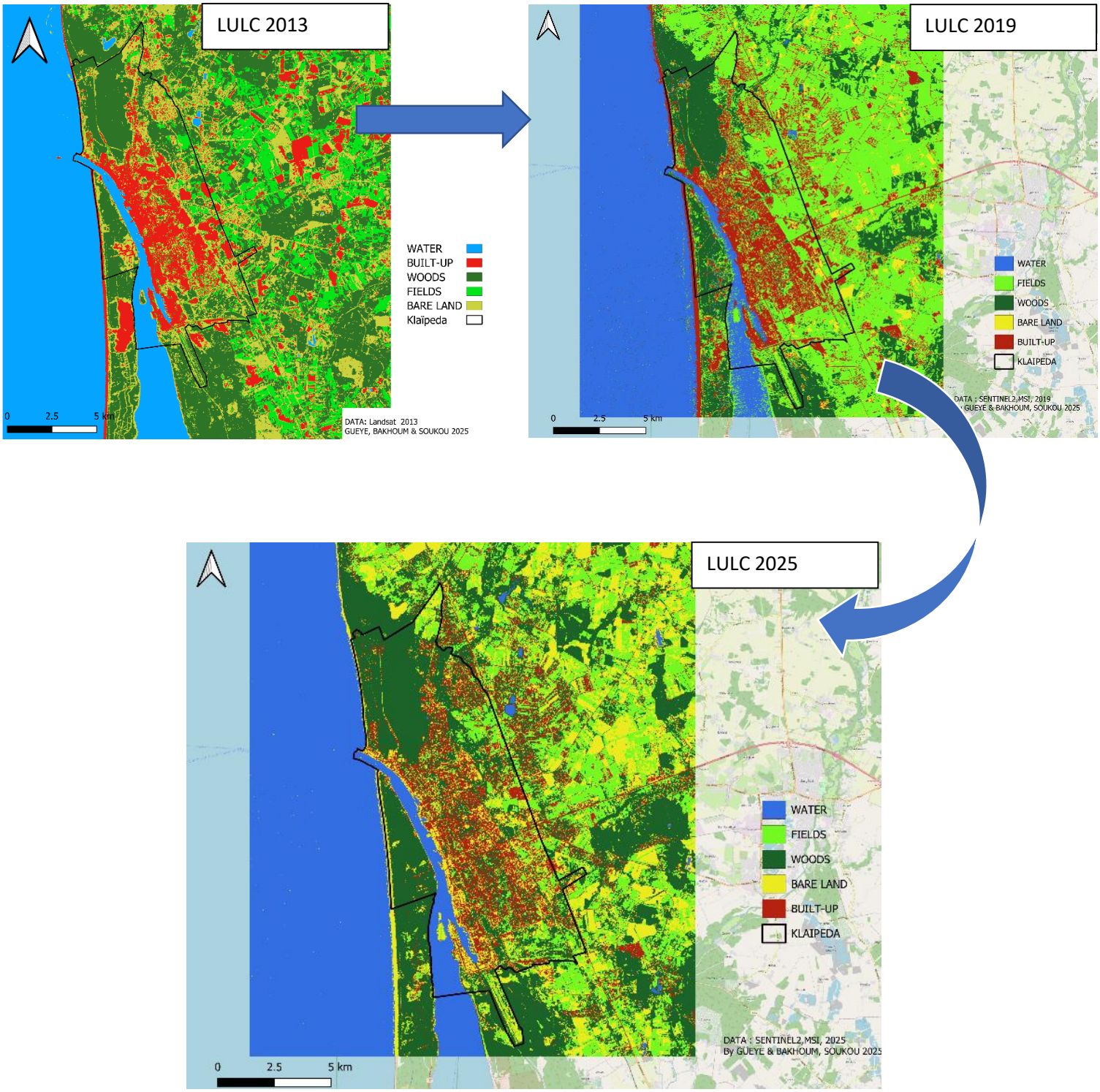
Figure 29: Land use land cover 2050



The prediction model illustrates the potential evolution of Land Use/Land Cover in the Klaipėda region. The map on the left represents the projected situation for the year 2025, while the cube on the right shows the simulated scenario for 2050.Between these two dates, a significant expansion of urbanized areas can be observed, reflecting the growth of artificial surfaces at the expense of natural and agricultural land. This trend indicates a likely urban expansion linked to demographic growth and economic development. At the same time, a reduction in vegetated areas appears, which may be associated with increasing anthropogenic pressure or land-use conversion processes. This model is therefore useful for anticipating future spatial transformations and serves as a decision-support tool for territorial planning, particularly in relation to sustainable land management and urban development.

GLOBAL RESULT: LULC CHANGE

Figure 30: GLOBAL RESULT: LULC CHANGE



General conclusion

This study provided a comprehensive analysis of urban and coastal land use and land cover (LULC) changes in the Klaipėda region using multi-sensor remote sensing data, including Landsat 8, Sentinel-2, and Kanopé V. By applying a structured methodology involving preprocessing, resampling, pansharpening, and combined supervised and unsupervised classification approaches, it was possible to generate LULC maps distinguishing five primary classes: water bodies, fields, wooded areas, bare soil, and built-up surfaces. Comparative analysis with reference datasets such as CORINE Land Cover (CLC), Urban Atlas, and OpenStreetMap (OSM) allowed for a detailed assessment of the detectability and spatial accuracy of urban objects across sensors.

The results highlight the strengths and limitations of each sensor. Landsat 8 provides consistent long-term coverage and broad temporal series, making it suitable for regional monitoring and trend analysis, though its coarser spatial resolution limits detailed urban object detection. Sentinel-2 offers improved spatial and spectral detail, enabling more accurate representation of medium-scale urban and vegetation features. Kanopé V, with its very high spatial resolution, allows for precise mapping of individual buildings and fine urban structures, providing the highest detectability among the three sensors.

Overall, the study demonstrates that combining multi-sensor datasets enhances our understanding of urban and coastal dynamics. It emphasizes the importance of selecting appropriate spatial and spectral resolutions according to study objectives and the type of land-cover objects of interest. Future work could focus on integrating additional high-resolution data, applying advanced machine learning classification methods, and developing automated workflows for continuous monitoring of urban growth and coastal changes in Klaipėda and similar regions.

In addition, the integration of the predictive model provides valuable insights into the potential future evolution of land use and land cover in the Klaipėda region. The projected scenarios for 2025 and 2050 reveal a likely expansion of built-up areas and a gradual reduction of natural vegetation, reflecting ongoing urbanisation pressures. These projections highlight the relevance of long-term spatial planning and support decision-making for sustainable coastal and urban management. By combining historical change detection with future modelling, this study offers a comprehensive perspective on past, present, and anticipated landscape dynamics.

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